iPSC editing with TALENs

Protocol by Ruilin Tian, Jason Hong and Martin Kampmann

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Reference:

Tian et al (2019). CRISPR Interference-Based Platform for Multimodal Genetic Screens in Human iPSC-Derived Neurons. Neuron pii: S0896-6273(19)30640-3. [Epub ahead of print] PubMed PMID: 31422865.

Reagents

Component	**Cat. No**	**Manufacturer**
Lipofectamine Stem	STEM00008	Invitrogen (Life Tech/Fisher)
E8 Medium	A1517001	Gibco (Life Tech/Fisher)
Stemflex Medium	A3349401	Gibco (Life Tech/Fisher)
Matrigel	356231	Corning (Discovery Labware)
Knockout DMEM	10829-018	Gibco (Life Tech/Fisher)
DPBS	14190144	D8537-24X500mL (Sigma)
Accutase	A11105-01	Gibco (Life Tech/Fisher)
Opti-Mem Reduced Serum	31985070	Gibco (Life Tech/Fisher)
Rock inhibitor (Y-27632)	1254/10	R&D Systems (Tocris/Biotechne)
TALEN-L (e.g. for CLYBL: pZT-C13-L1)	#62196	Addgene
TALEN-R (e.g. for CLYBL: pZT-C13-R1)	#62197	Addgene
BCL-XL plasmid	pEF1-BCL-XL-wpre-polyA P1102	Gift from Xiaobing Zhang, described in PMID: 30239926

1. Pre-coat 6 well plate
    a. Pre-coat 6 well plate with Matrigel (diluted 1:50 with Knockout DMEM) by adding 1mL to each well and let it sit in incubator for at least 20 minutes.

2. Passaging and Plating iPSC Cells
    a. Grow iPSCs until they are ~85% confluent in one well of a 6 well plate or larger format.

b. Remove old medium and wash them with DPBS 1x and add Accutase for 3-5 min to lift the cells.
c. Singularize the cells by gently pipetting them up and down several times.
d. In a 15mL conical tube, add DPBS and then add the lifted cells in Accutase.
e. Spin down at 200g for 5 min.
f. Aspirate supernatant and resuspend cells in Stemflex.
g. Perform cell count.
h. Remove the Matrigel in the pre-coated plate and add appropriate amount of Stemflex medium with Rock inhibitor (1000x)
i. Re-seed 0.5M cells into a 6 well plate so that it can reach ~60% confluency the next day.
j. Put plate in incubator and culture overnight.

3. Transfection
    a. The following day, change the media with new E8 media.
    b. Prepare the following mixes:
        i. Tube 1
            1. Opti-Mem - 100uL
            2. Lipofectamine Stem Reagent - 10uL
        ii. Tube 2
            1. Opti-Mem - 100uL
            2. DNA mix (0.5-5.0ug total)
                a. Optimized TALENS ratio
                    i. Your Plasmid of choice - 1.5ug
                    ii. TALENS L - 0.75ug
                    iii. TALENS R - 0.75ug
                    iv. BCL-XL - 0.3ug
    c. Add tube 2 to tube 1 and mix well.
    d. Incubate the mixture for 10 minutes.
    e. Add the whole mixture to your cells in the 6 well plate.
    f. Incubate overnight and change the media to Stemflex the next day.
    g. If confluent, passage and expand them into 10cm plate.
    h. When confluent in 10cm plate, continue to selection protocol.